Chuong Dang TA

Stockholm, Sweden | Willing to relocate

chuong.tadang.hust@gmail.com +46 769781960 linkedin.com/in/chuongta chuongta.github.io

About Me

Energy systems and data science enthusiast with a thermal engineering background. Expected to graduate in **September 2026** from an Erasmus Mundus MSc (**DENSYS**). For my Master's Thesis at **rebase.energy** (Stockholm), I forecasted wind turbine icing power losses using an extended **IEA Wind Task 19** framework and **LightGBM**. Active builder of energy data side projects, including probabilistic day-ahead price forecasting, smart heat pump control, and energy storage optimization. Published author in 2 peer-reviewed Q1 journals (IFs: 11.8, 9.2).

Education and Training

University of Lorraine, France & Politechnique of Turin, Italy - densys.univ-lorraine.fr *Sep 2024 – Sep 2026*
Master of Science in Decentralised and Smart Energy Systems (DENSYS)
A two-year Erasmus Mundus Joint Master's program (120 ECTS) focusing on the modeling, control, and optimization of decentralised and renewable energy systems.

Selected Coursework:
- Univeristy of Lorraine: [Data and Forecasting in Microgrids], [Optimal Local Design Energy Network].
- Politecnique of Turin: [Polygeneration and Advanced Energy Systems], [Smart Electricity Systems].

Key Projects:
- *Machine Learning (ML) Enhanced **LCA** of a North Sea Offshore Wind Farm*: Developed integrated LCA-ML framework achieving 12,600× computational acceleration (R^2=98.5%) for rapid environmental optimization across 285 design scenarios.

Hanoi University of Science and Technology (HUST), Vietnam - hust.edu.vn/en *Aug 2018 – Jan 2023*
Bachelor in Thermal Engineering
Graduated **1st/218** with a CGPA of 3.54/4.00 from Vietnam's leading technical university. Core studies covered thermodynamics, heat transfer, district heating and cooling systems, and power plant systems.
Bachelor Thesis: *"Techno-Economic Analysis of Internal Combustion Engines Using Diesel and LNG"* (Grade: 9.5/10)

Scientific Research - ORCID: 0009-0009-7530-7430

- Dang-Chuong Ta; Thanh-Hoang Le; Long Van Phan; Hoang-Luong Pham, "Feasibility Analysis of Hydrogen Co-Firing in Vietnam's Gas Power Plants for the Period 2035–2050," *Energy Conversion and Management*, Impact Factor: 11.8, **2025**. DOI: 10.1016/j.enconman.2025.120192
- Dang-Chuong Ta; Thanh-Hoang Le; Hoang-Luong Pham, "An Assessment of the Potential for Large-Scale Hydrogen Export from Vietnam to Asian Countries: Techno-Economic Analysis, Transport Options, and Energy Carrier Comparison," *International Journal of Hydrogen Energy*, Impact Factor: 9.2, **2024**. DOI: 10.1016/j.ijhydene.2024.04.033
- *(Ongoing)* Dang-Chuong Ta et al., "Wind Farm-Scale Forecasting of Turbine Icing Power Losses Using Multi-Ridge Regression Models," based on MSc thesis research at rebase.energy - extending the turbine-level classifier-regressor framework with multi-ridge regression models and upscaling to wind farm-level forecasts.

Work Experience

rebase.energy, Stockholm, Sweden *Jul 2026 – Present*
Junior Data Scientist - Wave Energy Converter Power Forecasting
- Developing power forecasting models for Wave Energy Converters, benchmarking multiple ridge-regularised and tree-based models (**Ridge Regression, Random Forest, LightGBM**), Probabilistic Forecasting Models (**Quantile Regression, Quantile Regression Forests, Boostrapped residuals**).

rebase.energy, Stockholm, Sweden *Feb 2026 – Jul 2026*
Master Thesis Student - Thesis Project Link
- **Icing Detection & Feature Engineering:** Established icing labels on multi-year Swedish turbine SCADA data using an extended **IEA Wind Task 19 Sigmoid Performance Ratio** framework. Diagnosed inter-annual non-stationarity and wind speed-temperature overlap between icing and non-icing regimes, and corrected SCADA-ERA5-Land biases via **Quantile Mapping** and **LightGBM**. Applied **Spearman** correlation and **SHAP** screening across multiple physical feature groups to remove unstable predictors.

- **Icing Power-Loss Forecasting:** Built a two-stage **LightGBM** classifier-regressor, trained with walk-forward cross-validation over multiple winters, to forecast wind power icing losses at **1-36 hour** lead times. Benchmarked Persistence, Oracle, Realistic NWP, and SCADA-only scenarios on a blind winter test set, showing local SCADA signals dominate near-term skill while NWP forecasts are essential for longer horizons - a key trade-off for operational resilience under changing conditions.

Selected Projects

- **Probabilistic Electricity Price Forecasting & Stochastic Storage Optimisation** - chuongta.github.io/EnergyForecasting
  Self-directed blog series on Danish DK1 day-ahead prices: probabilistic forecasting with Quantile Regression, Quantile Regression Forests, and bootstrapped residuals; the resulting quantile forecasts drive a two-stage stochastic battery dispatch model in Pyomo, benchmarked against naive and perfect-foresight baselines.
- **Smart Heat Pump Control: Reinforcement Learning & Digital Twins**
  Built a physical model of a residential heat pump and thermal storage tank, integrated a Gradient Boosting-based Digital Twin to correct physical state estimation discrepancies, and trained a PPO reinforcement learning agent to optimize electricity cost arbitrage.
- **Structure of Electricity Markets & EU/Nordic Market Design**
  Writing a 2-post series covering the fundamental structure of electricity markets, the European electricity market design, and the clearing mechanisms and deregulation of the Nordic electricity market.

Technical Skills

- **Machine Learning:** Supervised, unsupervised, and reinforcement learning; hybrid, physics-informed models for energy system forecasting and predictive maintenance.
- **Data Science & Optimisation**: Applied **MILP, MINLP, and heuristic algorithms** (e.g., genetic algorithms) in Pyomo to optimize renewable energy systems.
- **Technical Toolset:** Python (PyTorch, Scikit-learn, TensorFlow, Pyomo, PyWake), MATLAB/Simulink, Modelica, QBlade, HOMER Pro, GIS, Aspen Plus, Typst, LaTeX.

Honors & Awards

- **Erasmus Mundus scholarship** for DENSYS Joint Master Program
- **JNED Award** for research on Japan's nuclear facilities (2023)

Short Courses & Certifications

- **Self-Directed Learning — Energy Markets:** DTU course **Renewables in Electricity Markets (46755)** (via lecture recordings). Topics: market structures, pricing mechanisms, day-ahead clearing, intraday trading, balancing markets, ancillary services, stochastic bidding.
- Wind Energy - Technical University of Denmark (DTU)
- 4th Training Course on Nuclear Power Plant Technology - The International Nuclear Energy Development of Japan Co., Ltd (JINED) (studied in Vietnam and Japan)
- Optimisation with Python: Complete Pyomo with Bootcamp A-Z
- Hydro, Wind & Solar Power - École Polytechnique de Paris

Language

- Vietnamese (Native)
- English (IELTS 7.5 - Full Professional)
- German (A2), French (A1)

Referees

Prof. Fabrice Lemoine - Chair of the DENSYS Erasmus Mundus Joint Master Degree, Université de Lorraine

fabrice.lemoine@univ-lorraine.fr

Marta Gandiglio - Associate Professor, Department of Energy (DENERG), Politecnico di Torino

marta.gandiglio@polito.it